



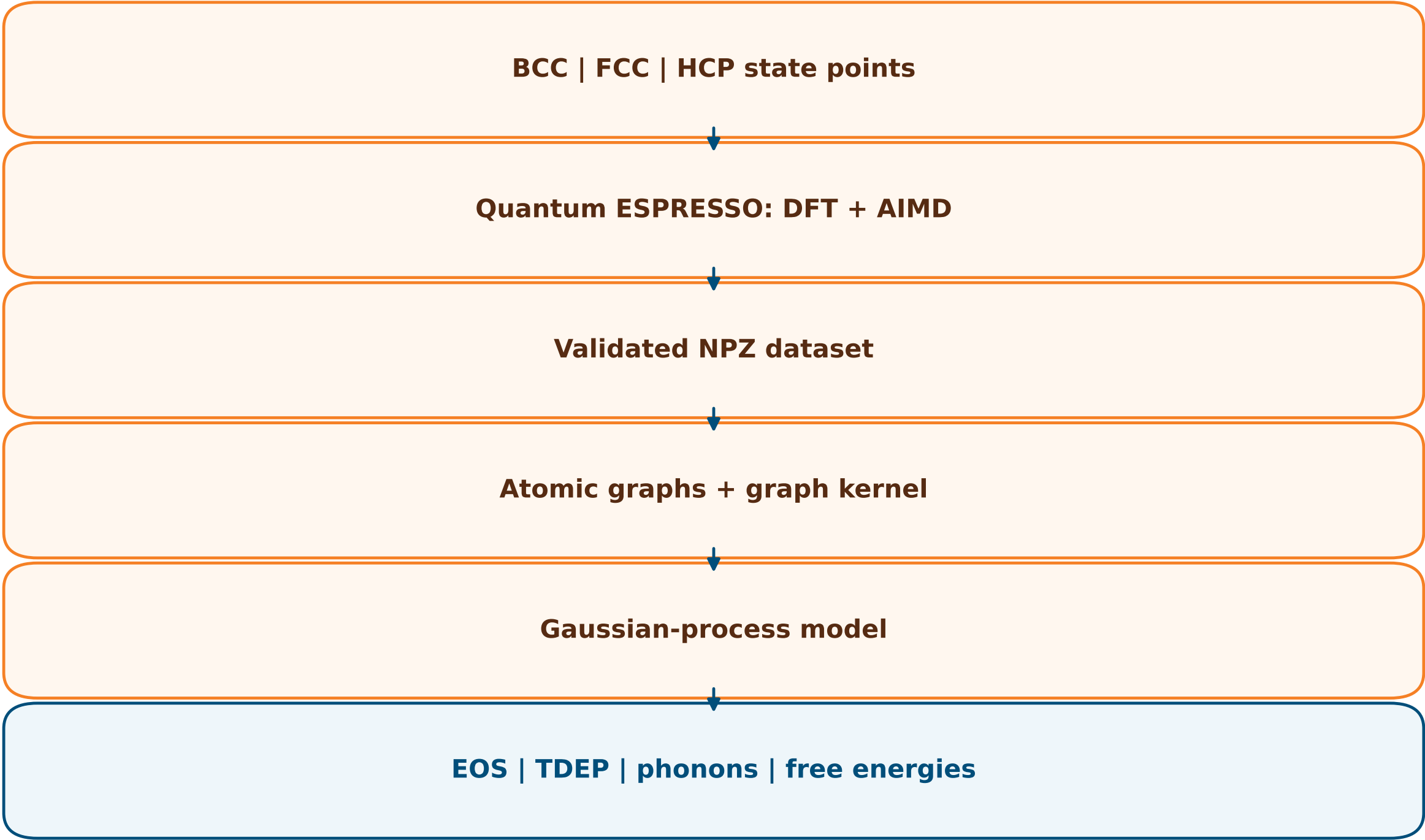
Earth's Core: Exploring Iron Allotropes under Extreme Conditions through First-Principles Simulations and Graph-Kernel-Based Machine Learning

SC26
RESEARCH
POSTERS

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INTRODUCTION

METHODOLOGY

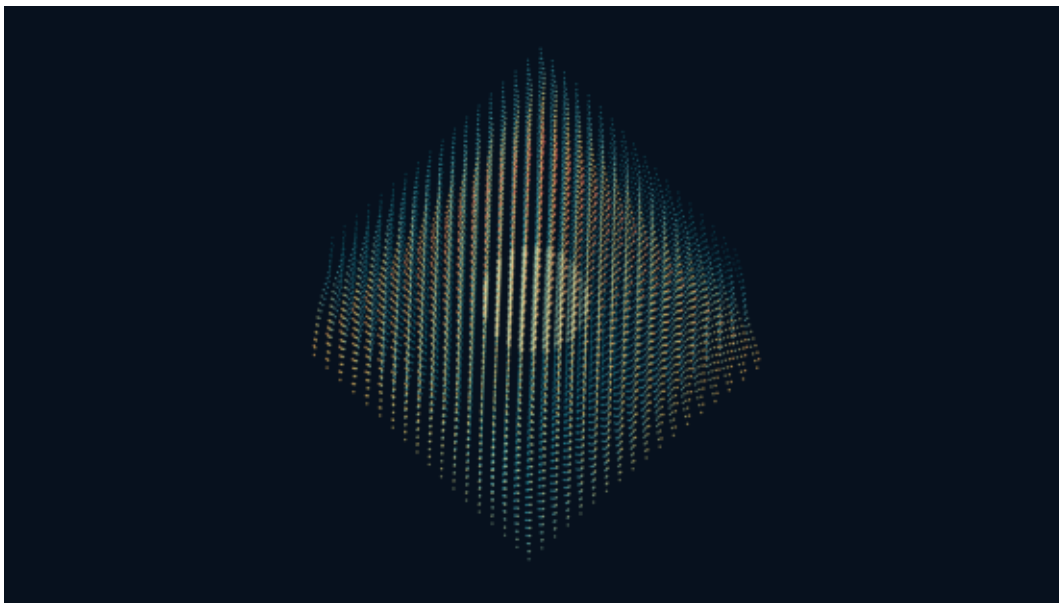


DISCUSSION

- Graph kernels provide a flexible representation across structurally diverse iron phases.
- Energy accuracy is encouraging, but force, EOS, phonon, and free-energy validation remain essential.
- Magnetic preparation changes individual branches and requires systematic convergence studies.

FUTURE WORK

- Train force components with trajectory-grouped splits.
- Expand magnetic sampling across volume and temperature.
- Compare BCC, FCC, and HCP free energies at matched conditions.
- Benchmark thousand-atom ML simulations and release reproducibility artifacts.



ACKNOWLEDGMENTS

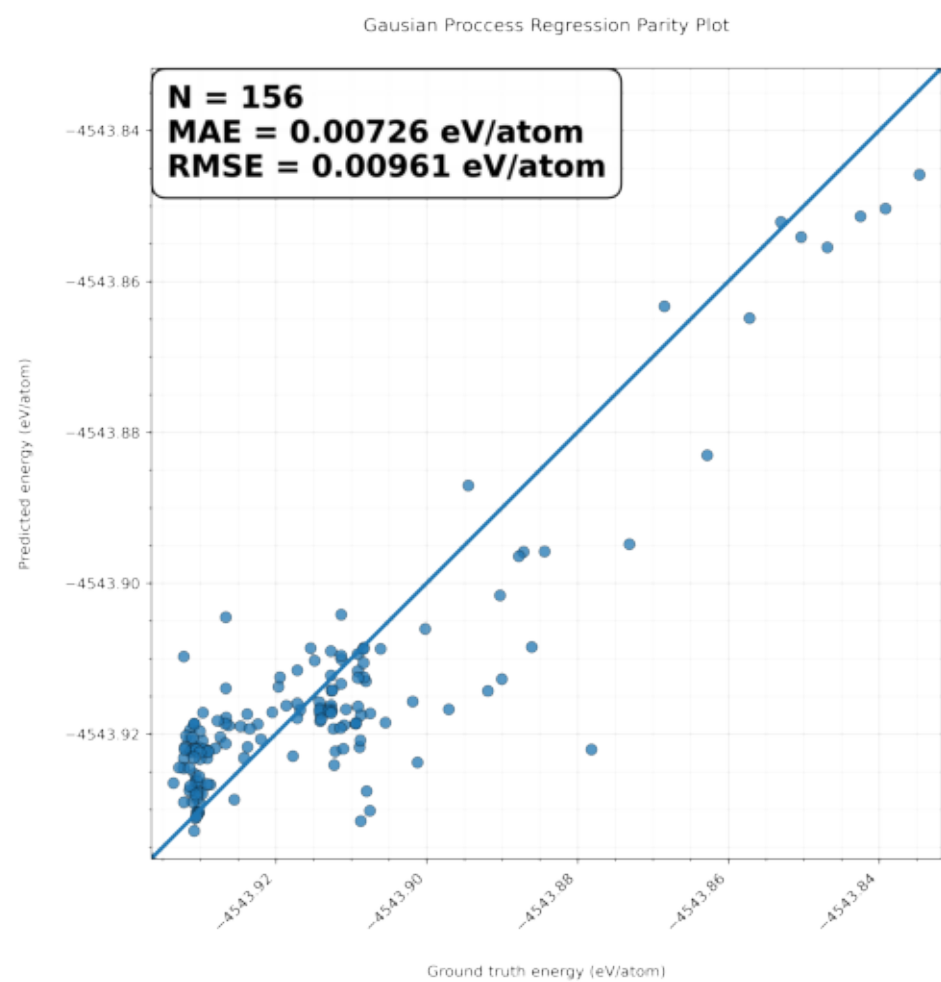
Computational resources, collaborators, funding sources, and award numbers will be added before final submission.

REFERENCES

- [1] Giannozzi et al., J. Phys.: Condens. Matter 21, 395502 (2009).
- [2] Hellman et al., Phys. Rev. B 84, 180301 (2011).
- [3] Vishwanathan et al., JMLR 11, 1201–1242 (2010).
- [4] Rasmussen and Williams, Gaussian Processes for Machine Learning (2006).

MOTIVATION

RESULTS



7.26

MAE (meV/atom)

9.61

RMSE (meV/atom)

156

evaluation structures

